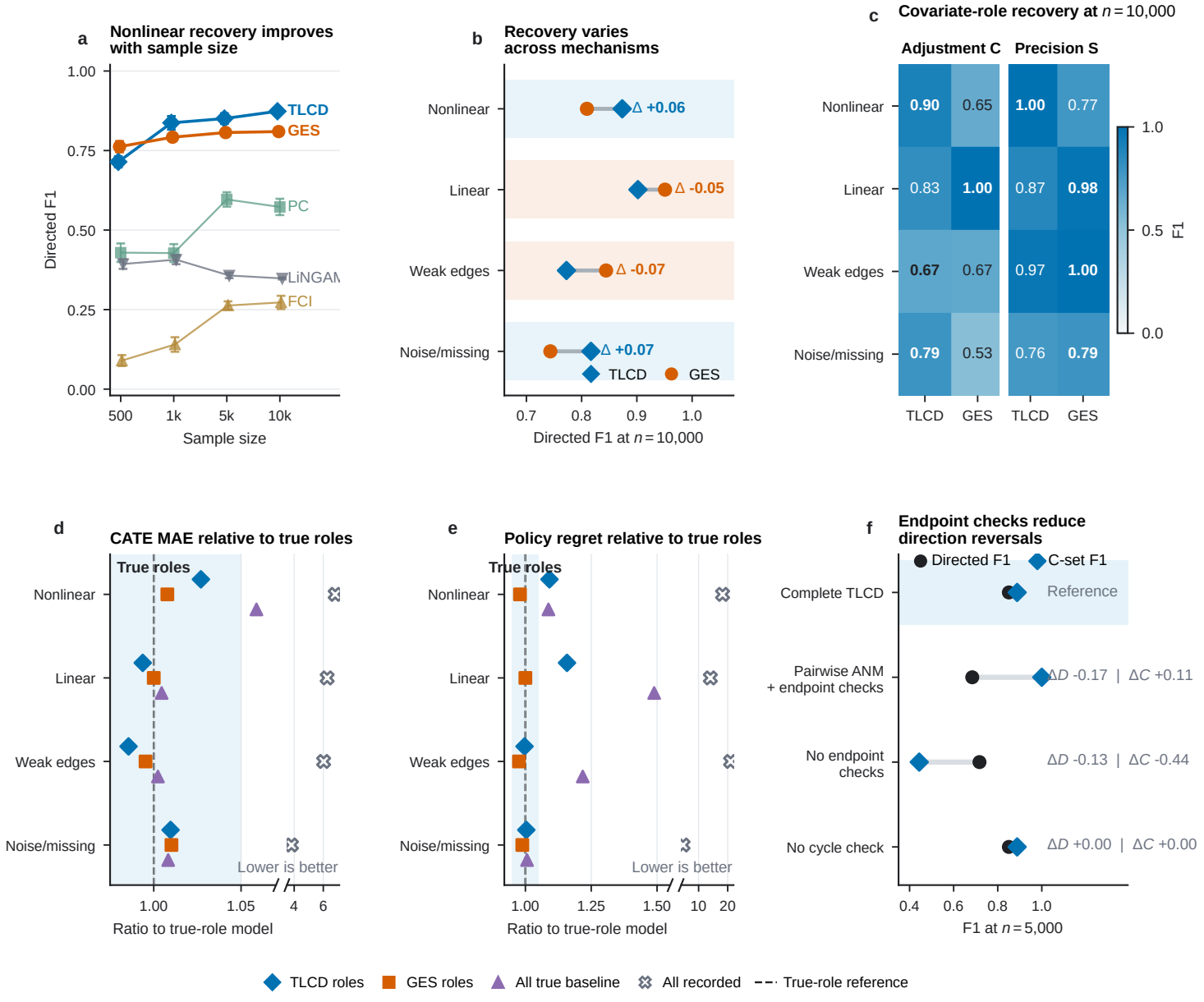
a
Nonlinear recovery improves with sample size
Directed F1
1.00
0.75
0.50
0.25
0.00
TLCD
GES
PC
LiNGAM
FCI
500
1k
5k
10k
Sample size
b
Recovery varies across mechanisms
Nonlinear
Linear
Weak edges
Noise/missing
Δ +0.06
Δ -0.05
Δ -0.07
Δ +0.07
TLCD
GES
0.7
0.8
0.9
1.0
Directed F1 at n = 10,000
c
Covariate-role recovery at n = 10,000
Adjustment C
Precision S
Nonlinear
Linear
Weak edges
Noise/missing
0.90
0.65
1.00
0.77
0.83
1.00
0.87
0.98
0.67
0.67
0.97
1.00
0.79
0.53
0.76
0.79
TLCD
GES
TLCD
GES
1.0
0.5
0.0
F1
d
CATE MAE relative to true roles
True roles
Nonlinear
Linear
Weak edges
Noise/missing
Lower is better
1.00
1.05
4
6
Ratio to true-role model
e
Policy regret relative to true roles
True roles
Nonlinear
Linear
Weak edges
Noise/missing
Lower is better
1.00
1.25
1.50
10
20
Ratio to true-role model
f
Endpoint checks reduce direction reversals
Directed F1
C-set F1
Complete TLCD
Reference
Pairwise ANM + endpoint checks
ΔD -0.17 | ΔC +0.11
No endpoint checks
ΔD -0.13 | ΔC -0.44
No cycle check
ΔD +0.00 | ΔC +0.00
0.4
0.6
0.8
1.0
F1 at n = 5,000
TLCD roles
GES roles
All true baseline
All recorded
True-role reference